Executive Summary

Testing Roughness in Realized Volatility

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1 Project question

Volatility is central in option pricing, hedging, and portfolio risk. Empirical volatility paths often look irregular, and this roughness is commonly summarized by a Hurst parameter H. Brownian-like roughness corresponds to $H \approx 0.5$, while rough-volatility models often use much smaller values such as $H \approx 0.1 - 0.3$.

The main question of this project is:

> Is the roughness observed in realized volatility a genuine feature of spot volatility, or can it be introduced by the way realized volatility is computed?

2 Methodology

We distinguish between spot volatility and realized volatility. In a log-price model

$$\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}B_t,$$

the process σ_t is the latent spot volatility. It is not directly observed from market prices. From observed log returns $\Delta X_i = X_{t_{i+1}} - X_{t_i}$, realized volatility over a window is computed by

$$\mathrm{RV}_{t,t+\Delta} = \left(\sum_{t_i \in [t,t+\Delta]} (\Delta X_i)^2 \right)^{1/2}.$$

Thus, realized volatility is an observable proxy constructed from the price path, but it is not the same object as σ_t.

To estimate roughness, we use the normalized p-variation statistic. For observations $X_0, \ldots, X_L$ on $[0,T]$, split the path into K coarse blocks and compute

$$W(p) = \sum_{i=0}^{K-1} \frac{|X_{b_{i+1}} - X_{b_i}|^p}{\sum_{j=b_i}^{b_{i+1}-1} |X_{j+1} - X_j|^p} (t_{b_{i+1}} - t_{b_i}).$$

Then choose

$$\widehat{p} \approx \operatorname*{arg\,min}_{p \in [1,12]} |W(p) - T|, \qquad \widehat{H} = \frac{1}{\widehat{p}}.$$

Smaller values of $\widehat{H}$ correspond to rougher paths.

3 Main experiments and findings

3.1 Fractional Brownian motion sanity check

The estimator was first tested on simulated fractional Brownian motion paths, where the true Hurst parameter is known. The estimator recovered the known roughness reasonably well. For example, for $H = 0.3$, the mean estimate over 50 paths was 0.2991, and for $H = 0.5$, the mean estimate was 0.5036.

3.2 Fractional OU stochastic-volatility simulation

We simulated

$$\mathrm{d}Y_t = -\gamma Y_t \,\mathrm{d}t + \theta \,\mathrm{d}B_t^H, \qquad \sigma_t = \sigma_0 e^{Y_t},$$

and then generated prices from

$$\mathrm{d}X_t = -\frac{1}{2}\sigma_t^2 \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t.$$

The estimator recovered the roughness of instantaneous volatility well: $\widehat{H}(\sigma) \approx H$. However, the realized-volatility estimates stayed below 0.3 across all tested values of H when the realized-volatility window was fixed at $m = 10$.

True H	$\widehat{H}(\sigma)$	$\widehat{H}(\mathrm{RV})$
0.10	0.088	0.224
0.20	0.211	0.218
0.30	0.293	0.201
0.40	0.392	0.145
0.50	0.499	0.166
0.60	0.597	0.164
0.70	0.683	0.163
0.80	0.773	0.158

3.3 Sensitivity to realized-volatility window size

For the fixed true roughness $H = 0.3$, the instantaneous-volatility estimate remained close to 0.3, while the realized-volatility estimate changed strongly with the window size.

RV window m	Window length Δ	$\widehat{H}(\sigma)$	$\widehat{H}(\mathrm{RV})$
10	0.00011	0.29056	0.18523
30	0.00033	0.28049	0.32981
100	0.00111	0.28381	0.49889
300	0.00333	0.30167	0.65755

This shows that $\widehat{H}(\mathrm{RV})$ is highly sensitive to the realized-volatility construction, even when the true volatility roughness is fixed.

4 Market-data evidence

We also applied the method to 1-minute AAPL prices downloaded with `yfinance`. The dataset contained 7410 one-minute price observations over 19 trading days, from 2026-06-10 to 2026-07-08. Realized volatility was computed using overlapping windows of different lengths.

RV window (minutes)	Length of RV path	$\widehat{H}(\mathrm{RV})$
2	7372	0.1132
5	7315	0.2168
10	7220	0.3267
15	7125	0.4258
30	6840	0.5025
60	6270	0.6313

The market-data experiment shows the same qualitative effect as the simulations: small windows give rougher estimates, while larger windows smooth the realized-volatility path and increase $\widehat{H}(\mathrm{RV})$.

Conclusion

The project shows that roughness estimated from realized volatility should be interpreted carefully. The estimator can recover known roughness when applied to simulated paths such as fractional Brownian motion or the true instantaneous volatility in a simulation. However, realized volatility is a constructed object, and its estimated roughness can change substantially with the realized-volatility window size.

Final note: A low value of $\widehat{H}(\mathrm{RV})$ should not automatically be interpreted as proof that spot volatility is rough. The realized-volatility window size and construction method must be reported and tested carefully.